Lesson notes · for you

# When AI gets it wrong

AI · Level 1 · Lesson 4

**Time**
26 min

**Materials**
A notebook page for your “AI mistake of the week”; A pen; A phone you can use (yours or a family member’s), if you have one

## Learning goals

- Tell a false alarm from a miss.
- Explain that people choose which mistake matters more, and that gaps in the data cause mistakes.
- Check, ask a person and report when AI gets it wrong.

## Activity: Be the filter

1. Warm-up (3 min): Each person shares one time an app got something wrong: a translation, a suggestion, a message in the wrong folder. Was it a false alarm or a miss?
2. Write (5 min): Together, write 10 short messages on slips of paper: 6 real ones (homework, a match moved, a family message) and 4 scams (a prize, free data, an urgent request for a PIN). Mark the answer on the back. Leave out real names and numbers.
3. Filter (8 min): Shuffle the slips. One person is the filter and calls each message “spam” or “safe” without seeing the back. The others check and count false alarms (real messages called spam) and misses (scams called safe).
4. Move the line (6 min): Play twice more with a new filter. First be very strict: call anything slightly odd “spam”. Then be very relaxed: call only obvious scams “spam”. Compare the counts: what happens to false alarms and misses?
5. Decide (4 min): Agree which mistake matters more for school messages, and why. Then say what you would do if a real school message went to spam: check, ask, report.

**You need 10 slips of paper and a pen. Write the answer (real or scam) on the back of each slip before you start, and use made-up names and numbers only.**

## Discussion

- For a school message filter, which mistake is worse: a false alarm or a miss?
- Whose voices or languages might an AI tool near you know least?
- Have you ever reported a mistake to an app? What happened?

## Check yourself

- Give one example of a false alarm and one of a miss.

**Try it this week**
This week, catch one AI mistake: a wrong suggestion, a bad translation or a message in the wrong folder. Write down whether it was a false alarm or a miss, and what might have caused it.

Worksheet

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Name _____ Class _____ Date _____

**1** **For a smoke alarm, which mistake is worse?**

◯ A false alarm from cooking steam ◯ Both are exactly as bad ◯ Missing a real fire

**2** **The computer decides by itself which mistake matters more.**

◯ True ◯ False

**3** **Which of these are misses? Choose all that are.**

☐ A scam lands in your inbox with no warning
☐ A spell-checker lets a misspelt word through
☐ A smoke alarm stays quiet during a real fire
☐ Your friend's message is sent to spam
☐ A smoke alarm beeps at cooking steam

**4** **Match each mistake to its most likely cause.**

| | |
|---|---|
| After an update, many real school messages go to spam | Dim light changed the numbers |
| A translator mixes up words in your grandmother's language | The line was moved too low |
| Face unlock fails in a dark room | It was set to go off easily, on purpose |
| A smoke alarm beeps whenever someone fries eggs | Little of that language in its examples |

**Think of an AI mistake you have seen. Was it a false alarm or a miss, and who could it harm?**

Check your answers

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**1** **Missing a real fire**

A beep from steam is annoying, but a missed fire can cost lives.

**2** **False**

People choose, by asking who is hurt by each mistake and how badly.

**3** **A scam lands in your inbox with no warning; A smoke alarm stays quiet during a real fire; A spell-checker lets a misspelt word through**

In a miss, something real gets through with no alarm: the scam, the fire, the misspelt word.

**4** **After an update, many real school messages go to spam — The line was moved too low**
**A translator mixes up words in your grandmother's language — Little of that language in its examples**
**Face unlock fails in a dark room — Dim light changed the numbers**
**A smoke alarm beeps whenever someone fries eggs — It was set to go off easily, on purpose**

Each mistake has its own cause. A threshold moved too low flags real messages; a data gap leaves a language poorly learned; dim light changes every pixel's number.